

SIMATS ENGINEERING

ASSESSMENT TOOL 3

**COURSE NAME: Artificial Intelligence
CSA17**

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TECHNICAL PRESENTATION ASSESSMENT REPORT

1. Reactive Agent-Based Traffic Control System

Introduction

Modern cities face severe congestion due to increasing vehicle density. Traditional fixed-time traffic signals fail to adapt to dynamic road conditions. A **Reactive Agent** provides an intelligent, real-time solution by responding instantly to environmental inputs.

System Overview

A Reactive Agent operates without long-term planning. It continuously senses traffic conditions and immediately adjusts signal timings to reduce congestion.

Architecture

1. Sensing Layer

- IoT vehicle counters
- CCTV cameras with object detection
- GPS data from connected vehicles
- Roadside units (RSUs)

2. Streaming Data Processing

Real-time pipelines compute:

- Vehicle density
- Queue length
- Waiting time
- Emergency vehicle detection

3. Reactive Agent Logic

Perception: Reads current traffic metrics every few seconds.

Decision Rules:

- Extend green if queue length exceeds threshold
- Shorten green if lane is free
- Override cycle for emergency vehicles
- Allocate pedestrian crossing when requested

Action: Updates traffic signal phases dynamically.

Outcome

- Reduced congestion
- Faster lane clearance
- Priority for emergency vehicles
- Adaptive traffic management

2. Goal-Based Agent for Healthcare Diagnosis

Introduction

Healthcare diagnosis requires structured reasoning, accurate interpretation of patient data, and optimal treatment selection. A **Goal-Based Agent** achieves this by setting diagnostic goals and choosing actions that fulfill them.

System Overview

The agent aims to:

- Identify the disease
- Recommend the best treatment

- Minimize risk and side effects

Architecture

1. Data Pipeline

- Patient history
- Lab results
- Medical imaging
- Vital signs
- Wearable device data

2. Goal Formulation

Primary goals:

- G1: Diagnose the condition
- G2: Select optimal treatment
- G3: Ensure patient safety

3. Decision-Making Logic

- Rule-based clinical guidelines
- Decision trees for symptom analysis
- Probabilistic models (Bayesian networks)
- Risk scoring algorithms

4. Treatment Pathway Selection

Agent evaluates:

- Drug effectiveness
- Side effects
- Patient allergies
- Comorbidities
- Cost and availability

Outcome

- Faster diagnosis
- Personalized treatment

- Reduced medical errors
- Evidence-based decision-making

3. Utility-Based Agent for E-Commerce Recommendations

Introduction

E-commerce platforms must deliver personalized recommendations to improve user satisfaction and business revenue. A **Utility-Based Agent** selects products that maximize a defined utility function.

System Overview

Utility represents:

- Relevance
- Satisfaction
- Price attractiveness
- Novelty
- Business value

Architecture

1. Data Sources

- Browsing history
- Purchase records
- Ratings and reviews
- Product metadata
- Price and discount information

2. Distributed Processing

Using large-scale data systems:

- Compute relevance scores
- Analyze user behavior
- Cluster similar users
- Predict purchase likelihood

3. Utility Function

$$U(i, temuser) = w_1R + w_2S + w_3P + w_4N$$

4. Agent Behavior

- Assigns utility to each product
- Ranks items based on utility
- Recommends top-N items
- Learns from feedback
- Adjusts weights dynamically

Outcome

- Highly personalized recommendations
- Increased conversion rates
- Improved user satisfaction

4. Learning Agent for Personalized Education System

Introduction

Students learn at different speeds and have unique strengths. A **Learning Agent** adapts content difficulty and learning paths using feedback and performance data.

System Overview

The agent continuously improves its teaching strategy by learning from student interactions.

Architecture

1. Data Collection

- Quiz scores
- Time spent on lessons
- Mistake patterns
- Engagement metrics
- Content usage logs

2. Distributed Processing

Large-scale analytics detect:

- Weak topics
- Learning patterns
- Dropout risk
- Preferred content formats

3. Agent Components

- **Performance Element:** Chooses next content
- **Learning Element:** Updates internal model
- **Critic:** Evaluates outcomes
- **Feedback Loop:** Reinforces successful strategies

4. Adaptive Behavior

- Increases difficulty when student performs well
- Provides remedial content when struggling
- Recommends personalized learning paths
- Learns best teaching strategies

Outcome

- Improved learning outcomes
- Higher engagement
- Personalized education at scale

5. Multi-Agent System for Disaster Management

Introduction

Disaster management requires fast detection, accurate risk assessment, and coordinated response. A **Multi-Agent System (MAS)** provides decentralized intelligence for handling complex disaster scenarios.

System Overview

Multiple agents collaborate to detect disasters and coordinate response strategies.

Agent Types

1. Sensor Agents

- Collect IoT sensor data
- Monitor water levels, tremors, weather
- Analyze satellite images
- Track social media signals

2. Decision Agents

- Fuse multi-source data
- Compute risk scores
- Predict disaster severity
- Trigger alerts

3. Communication Agents

- Send warnings to authorities
- Coordinate evacuation routes
- Share updates across regions

Coordination Strategies

- **Blackboard system:** Shared knowledge base
- **Message passing:** Agents exchange states
- **Distributed decision-making:** Local + global coordination
- **Real-time updates:** Continuous monitoring

Outcome

- Early disaster detection
- Faster response
- Reduced casualties
- Efficient resource allocation

Conclusion

This technical presentation assessment demonstrates the design of five intelligent agent-based systems across critical domains: traffic control, healthcare, e-commerce,

education, and disaster management. Each system integrates advanced agent architectures with real-time data processing, distributed analytics, and adaptive decision-making. These designs reflect modern AI engineering practices and showcase how intelligent agents can transform large-scale, real-world applications.